1 Malik is playing a game in which he has to throw a 6 on a fair six-sided die to start the game. Find the probability that

(i) Malik throws a 6 for the first time on his third attempt, **[3]**

(ii) Malik needs at most ten attempts to throw a 6. **[2]**

2 The probability distribution of the random variable X is given by the formula

$$P(X = r) = k(r^2 - 1) \text{ for } r = 2, 3, 4, 5.$$

(i) Show the probability distribution in a table, and find the value of k. **[3]**

(ii) Find $E(X)$ and $Var(X)$. **[5]**

3 The weights of bags of a particular brand of flour are quoted as 1.5 kg. In fact, on average 10% of bags are underweight.

(i) Find the probability that, in a random sample of 50 bags, there are exactly 5 bags which are underweight. **[3]**

(ii) Bags are randomly chosen and packed into boxes of 20. Find the probability that there is at least one underweight bag in a box. **[2]**

(iii) A crate contains 48 boxes. Find the expected number of boxes in the crate which contain at least one underweight bag. **[2]**

4 Every evening, 5 men and 5 women are chosen to take part in a phone-in competition. Of these 10 people, exactly 3 will win a prize. These 3 prize-winners are chosen at random.

(i) Find the probability that, on a particular evening, 2 of the prize-winners are women and the other is a man. Give your answer as a fraction in its simplest form.

[4]

(ii) Four evenings are selected at random. Find the probability that, on at least three of the four evenings, 2 of the prize-winners are women and the other is a man.

[4]

5 A rugby team of 15 people is to be selected from a squad of 25 players.

(i) How many different teams are possible?

[2]

(ii) In fact the team has to consist of 8 forwards and 7 backs. If 13 of the squad are forwards and the other 12 are backs, how many different teams are now possible?

[2]

(iii) Find the probability that, if the team is selected at random from the squad of 25 players, it contains the correct numbers of forwards and backs.

[2]

6 The amount of data, X megabytes, arriving at an internet server per second during the afternoon is modelled by the Normal distribution with mean 435 and standard deviation 30.

(i) Find

A $P(X < 450)$,

[3]

B $P(400 < X < 450)$.

[3]

(ii) Find the probability that, during 5 randomly selected seconds, the amounts of data arriving are all between 400 and 450 megabytes.

[2]

The amount of data, Y megabytes, arriving at the server during the evening is modelled by the Normal distribution with mean μ and standard deviation σ.

(iii) Given that $P(Y < 350) = 0.2$ and $P(Y > 390) = 0.1$, find the values of μ and σ.

[5]

(iv) Find values of a and b for which $P(a < Y < b) = 0.95$.

[4]

7 The scores, X, in Paper 1 of an English examination have an underlying Normal distribution with mean 76 and standard deviation 12. The scores are reported as integer marks. So, for example, a score for which $75.5 \leqslant X < 76.5$ is reported as 76 marks.

(i) Find the probability that a candidate's reported mark is 76. **[4]**

(ii) Find the probability that a candidate's reported mark is at least 80. **[3]**

(iii) Three candidates are chosen at random. Find the probability that exactly one of these three candidates' reported marks is at least 80. **[2]**

The proportion of candidates who receive an A* grade (the highest grade) must not exceed 10% but should be as close as possible to 10%.

(iv) Find the lowest reported mark that should be awarded an A* grade. **[5]**

The scores in Paper 2 of the examination have an underlying Normal distribution with mean μ and standard deviation 12.

(v) Given that 20% of candidates receive a reported mark of 50 or less, find the value of μ. **[4]**

8 The wing lengths of native English male blackbirds, measured in mm, are Normally distributed with mean 130.5 and variance 11.84.

(i) Find the probability that a randomly selected native English male blackbird has a wing length greater than 135 mm.

[3]

(ii) Given that 1% of native English male blackbirds have wing length more than *k* mm, find the value of *k*.

[3]

(iii) Find the probability that a randomly selected native English male blackbird has a wing length which is 131 mm correct to the nearest millimetre.

[3]

It is suspected that Scandinavian male blackbirds have, on average, longer wings than native English male blackbirds. A random sample of 20 Scandinavian male blackbirds has mean wing length 132.4 mm. You may assume that wing lengths in this population are Normally distributed with variance 11.84 mm^2.

(iv) Carry out an appropriate hypothesis test, at the 5% significance level.

[8]

(v) Discuss briefly one advantage and one disadvantage of using a 10% significance level rather than a 5% significance level in hypothesis testing in general.

[2]

9 The random variable X represents the weight in kg of a randomly selected male dog of a particular breed. X is Normally distributed with mean 30.7 and standard deviation 3.5.

(i) Find

A P (X < 30),

[3]

B P(25 < X < 35).

[3]

(ii) Five of these dogs are chosen at random. Find the probability that each of them weighs at least 30 kg.

[2]

(iii) The weights of females of the same breed of dog are Normally distributed with mean 26.8 kg. Given that 5% of female dogs of this breed weigh more than 30 kg, find the standard deviation of their weights.

[4]

(iv) Sketch the distributions of the weights of male and female dogs of this breed on a single diagram.

[4]

10 Many types of computer have cooling fans. The random variable X represents the lifetime in hours of a particular model of cooling fan. X is Normally distributed with mean 50 600 and standard deviation 3400.

(i) Find $P(50000 < X < 55000)$.

[3]

(ii) The manufacturers claim that at least 95% of these fans last longer than 45000 hours. Is this claim valid?

[3]

(iii) Find the value of h for which 99.9% of these fans last h hours or more.

[3]

(iv) The random variable Y represents the lifetime in hours of a different model of cooling fan. Y is Normally distributed with mean μ and standard deviation σ. It is known that $P(Y < 60000) = 0.6$ and $P(Y > 50000) = 0.9$. Find the values of μ and σ.

[5]

(v) Sketch the distributions of lifetimes for both types of cooling fan on a single diagram.

[4]

11 In a particular country, 8% of the population has blue eyes. Find the probability that, in a random sample of 20 people, exactly two have blue eyes. [2]

12 Each day, for many years, the maximum temperature in degrees Celsius at a particular location is recorded. The maximum temperatures for days in October can be modelled by a Normal distribution; the appropriate Normal curve is shown in Fig. 6.

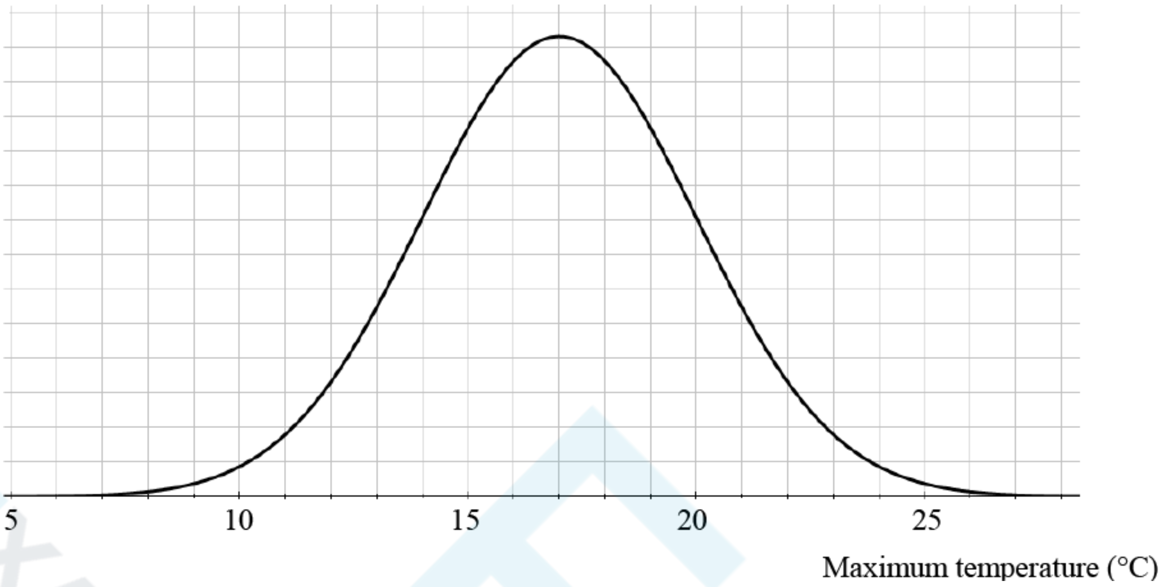


Fig. 6

(a)

(i) Use the model to write down the mean of the maximum temperatures.

(ii) Explain why the curve indicates that the standard deviation is approximately 3 degrees Celsius. [2]

Temperatures can be converted from Celsius to Fahrenheit using the formula $F = 1.8C + 32$, where F is the temperature in degrees Fahrenheit and C is the temperature in degrees Celsius.

(b) For maximum temperature in October in degrees Fahrenheit, estimate

- the mean,
- the standard deviation. **[2]**

13 The quality control department of a battery manufacturing company checks the lifetimes of the batteries produced by the company. The lifetimes, x minutes, for a random sample of 80 'Superstrength' batteries are shown in the table below.

Lifetime	$160 \le x < 165$	$165 \le x < 168$	$168 \le x < 170$	$170 \le x < 172$	$172 \le x < 175$	$175 \le x < 180$
Frequency	5	14	20	21	16	4

(a) Estimate the proportion of these batteries which have a lifetime of at least 174.0 minutes. **[2]**

(b) Use the data in the table to estimate

- the sample mean,
- the sample standard deviation. **[3]**

The data in the table on the previous page are represented in the following histogram:

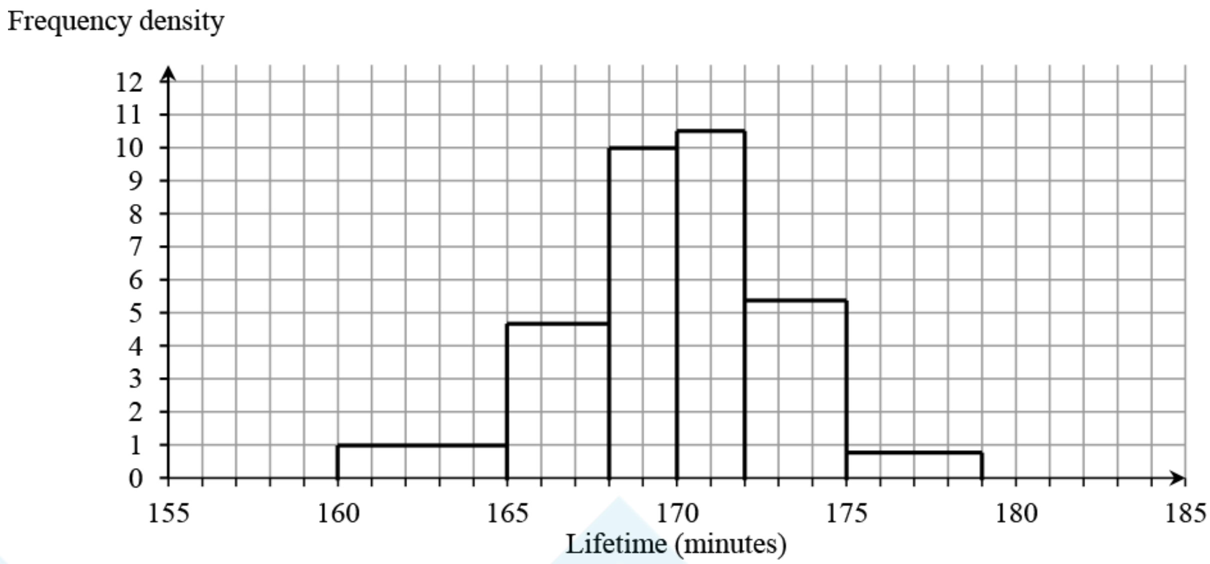


Fig. 15

A quality control manager models the data by a Normal distribution with the mean and standard deviation you calculated in part **(b)**.

(c) Comment briefly on whether the histogram supports this choice of model. **[2]**

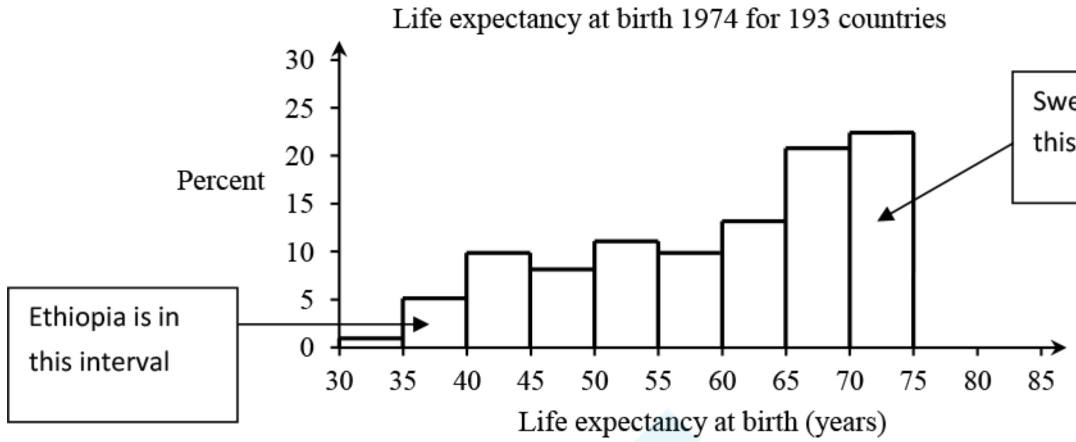
(d) (i) Use this model to estimate the probability that a randomly selected battery will have a lifetime of more than 174.0 minutes.

(ii) Compare your answer with your answer to part **(a)**. **[3]**

The company also manufactures ‘Ultrapower’ batteries, which are stated to have a mean lifetime of 210 minutes.

(e) A random sample of 8 Ultrapower batteries is selected. The mean lifetime of these batteries is 207.3 minutes. Carry out a hypothesis test at the 5% level to investigate whether the mean lifetime is as high as stated. You should use the following hypotheses $H_0 : \mu = 210$, $H1 : \mu < 210$, where μ represents the population mean for Ultrapower batteries. You should assume that the population is Normally distributed with standard deviation 3.4. **[5]**

14 Fig. 16.1, Fig. 16.2 and Fig. 16.3 show some data about life expectancy, including some from the pre-release data set (see http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls).



A book

Fig. 16.1

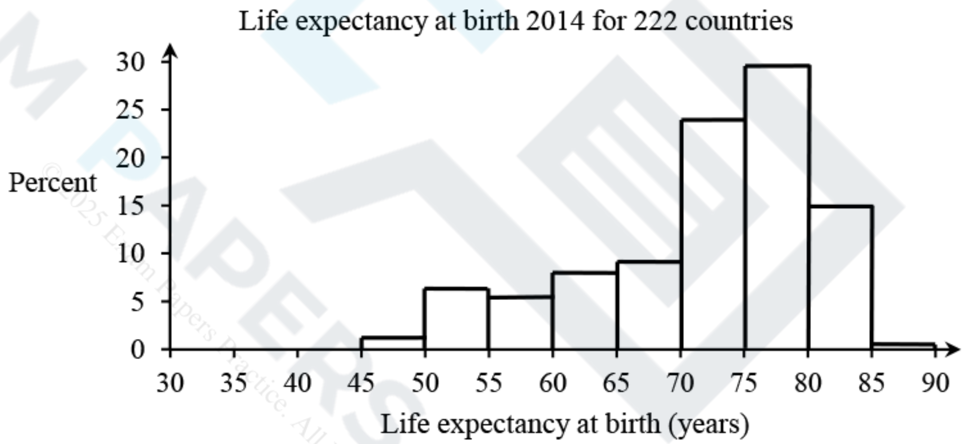


Fig. 16.2

Increase in life expectancy from 1974 to 2014 (years)

Source: CIA World Factbook and Gapminder

Increase in life expectancy for 193 countries from 1974 to 2014	
Number of values	193
Minimum	- 4.618
Lower quartile	6.9576
Median	9.986
Upper quartile	15.873
Maximum	30.742

Fig. 16.3

(a) Comment on the shapes of the distributions of life expectancy at birth in 2014 and 1974. **[2]**

(b) (i) The minimum value shown in the box plot is negative. What does a negative value indicate? **[1]**

(ii) What feature of Fig 16.3 suggests that a Normal distribution would **not** be an appropriate model for increase in life expectancy from one year to another year? **[1]**

(iii) Software has been used to obtain the values in the table in Fig. 16.3. Decide whether the level of accuracy is appropriate. Justify your answer. **[1]**

(iv) John claims that for half the people in the world their life expectancy has improved by 10 years or more. Explain why Fig. 16.3 does not provide conclusive evidence for John's claim. **[1]**

(c) Decide whether the maximum increase in life expectancy from 1974 to 2014 is an outlier. Justify your answer. **[3]**

Here is some further information from the pre-release data set (see http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls).

Country	Life expectancy at birth in 2014
Ethiopia	60.8
Sweden	81.9

(d) (i) Estimate the change in life expectancy at birth for Ethiopia between 1974 and 2014.

(ii) Estimate the change in life expectancy at birth for Sweden between 1974 and 2014.

(iii) Give **one** possible reason why the answers to parts (i) and (ii) are so different. **[4]**

Fig.16.4 shows the relationship between life expectancy at birth in 2014 and 1974.

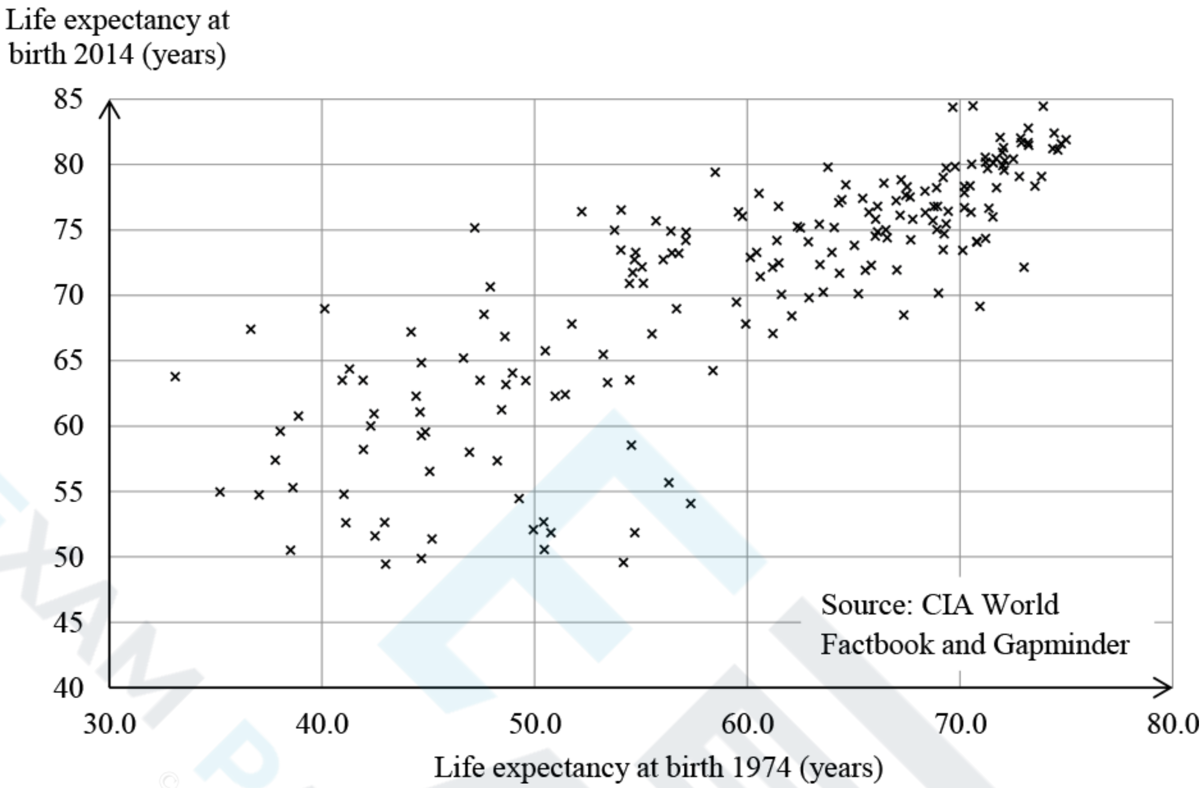


Fig. 16.4

A spreadsheet gives the following linear model for all the data in Fig 16.4.

(Life expectancy at birth 2014) = 30.98 + 0.67 × (Life expectancy at birth 1974)

The life expectancy at birth in 1974 for the region that now constitutes the country of South Sudan was 37.4 years. The value for this country in 2014 is not available.

(e) (i) Use the linear model to estimate the life expectancy at birth in 2014 for South Sudan. **[2]**

(ii) Give two reasons why your answer to part (*i*) is not likely to be an accurate estimate for the life expectancy at birth in 2014 for South Sudan. You should refer to **both** information from Fig 16.4 **and** your knowledge of the large data set. **[2]**

(f) In how many of the countries represented in Fig. 16.4 did life expectancy drop between 1974 and 2014? Justify your answer. **[3]**

15 A company operates trains. The company claims that 92% of its trains arrive on time. You should assume that in a random sample of trains, they arrive on time independently of each other.

(a) Assuming that 92% of the company's trains arrive on time, find the probability that in a random sample of 30 trains operated by this company

(i) exactly 28 trains arrive on time, **[2]**

(ii) more than 27 trains arrive on time. **[2]**

A journalist believes that the percentage of trains operated by this company which arrive on time is lower than 92%.

(b) To investigate the journalist's belief a hypothesis test will be carried out at the 1% significance level. A random sample of 18 trains is selected. For this hypothesis test,

- state the hypotheses,
- find the critical region. **[5]**

16 The discrete random variable X takes the values 0, 1, 2 and 3 only. You are given that

$$P(X = r) = k\,r!, \text{ where } k \text{ is a positive constant.}$$

(a) Show that $k = 0.1$. **[2]**

(b) Find $P(X = 3)$. **[1]**

(c) Calculate the probability of obtaining the value 3 exactly 16 times when 32 independent observations of X are made. **[2]**

17 The screenshot in Fig. 8 shows the probability distribution for the discrete random variable X, where $X \sim B(20, 0.9)$.

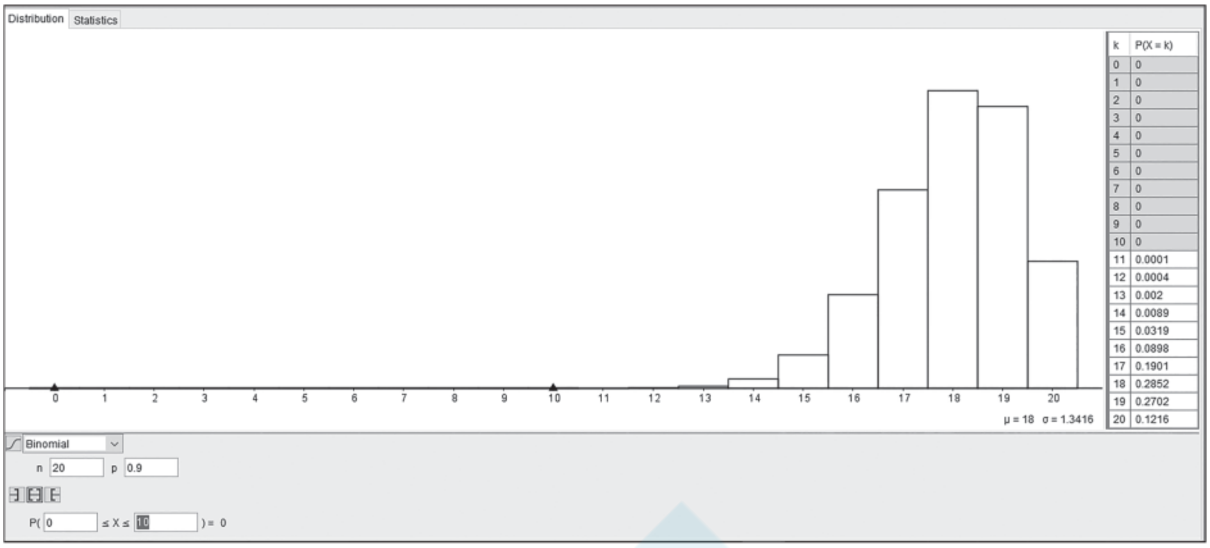


Fig. 8

(a) Describe the shape of the distribution. **[1]**

(b) Explain why the values of P($X = k$) for k = 0 to 10 inclusive are recorded as 0 in the table in the screenshot. **[1]**

(c) State which of the values from 0 to 20 is

(i) the most likely value of X, **[1]**

(ii) the least likely value of X. **[1]**

18 Every evening at bedtime my cat Arthur decides whether to spend the night inside or outside the house.

If Arthur spends a night inside, the probability that he will spend the next night outside is 0.6.

If Arthur spends a night outside, the probability that he will spend the next night inside is 0.9.

Arthur spends a Saturday night inside.

(a) Calculate the probability that Arthur will decide to spend the next Monday night outside. **[3]**

(b) Show that it is extremely likely that Arthur will spend at least one out of the next seven nights (Sunday to Saturday) inside. **[3]**

19 **(a)** The response time to a roadside callout from a breakdown recovery firm may be modelled by a Normal distribution. Over a long period it has been noted that 8% of the response times are less than 58 minutes and 19% of the response times are greater than 86 minutes. Determine estimates of the mean and standard deviation of the response time. **[7]**

(b) The recovery firm claim that 95% of their response times are less than 90 minutes. Investigate this claim. **[2]**

20 Between the islands of Tenerife and La Gomera there is a resident population of pilot whales. Studies in the past showed that 20% of the adult population were male. Due to a change in environmental conditions, it is thought that the proportion of males has decreased.

In order to investigate this, scientists caught and released a random sample of 43 different adult pilot whales. Exactly 3 of these whales were found to be male.

(a) Carry out a hypothesis test at the 5% level to investigate whether there is any evidence that the proportion of males has decreased. **[6]**

Previous studies also showed that the mean length of adult females was 2.98 metres. On another occasion the scientists caught a random sample of 39 different adult female pilot whales; the length, x metres, of each whale was measured before it was released. The data are summarized below.

$$\sum x = 120.7 \text{ and } \sum x^2 = 384.75$$

(b) Carry out a hypothesis test at the 5% level to investigate whether there is any evidence that the mean length of adult females has changed. **[8]**

21 The probability distribution for the discrete random variable X is shown in Fig. 3.

x	0	1	2	3	4
$P(X = x)$	$0.4a$	$0.5a$	$0.3a$	$0.2a$	$0.1a$

Fig. 3

(a) Calculate the value of the constant a. **[2]**

(b) Calculate $P(X \geq 2)$. **[2]**

(c) Describe the shape of the distribution. **[1]**

22 The pre-release data set consists of a subset (see http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls) of the data about countries from the CIA World Facebook.

Data for Croatia shows that the average number of mobile phone subscribers per 1000 population is 1111.72.

(a) Explain how this average can be greater than 1000. **[1]**

It is assumed that, once the data have been cleaned, the average number of phones per 1000 population in different countries can reasonably be approximated by a Normal distribution with mean 996 and standard deviation 407.

(b) State two ways in which the data may have been cleaned. **[2]**

Fig. 11.1

Fig. 11.1 shows a box plot of the cleaned data.

(c) Give the name of the measure which the arrow is pointing to. **[2]**

(d) Use the approximate Normal distribution to estimate a value for this measure. **[2]**

Fatima thinks that the mean number of mobile phone subscribers per 1000 population for the countries of the world has increased since the data were collected. In order to test this he obtains some up-to-date data for a random sample of countries and uses software to conduct the appropriate hypothesis test at the 5% level of significance. The output from the software is shown in Fig. 11.2.

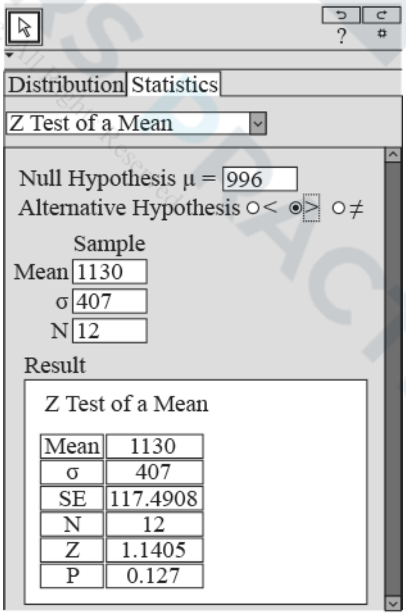


Fig. 11.2

(e) State the conclusion of Fatima's test, explaining your reasoning. **[4]**

(f) With reference to the pre-release data set, comment on the sample size Fatima has chosen. **[1]**

23 The probability distribution of the discrete random variable X is given in Fig. 4.

r	0	1	2	3	4
$P(X = r)$	0.2	0.15	0.3	k	0.25

Fig. 4

(a) Find the value of k. **[2]**

X_1 and X_2 are two independent values of X.

(b) Find $P(X_1 + X_2 = 6)$. **[3]**

24 **In this question you must show detailed reasoning.**

Research showed that in May 2017 62% of adults over 65 years of age in the UK used a certain online social media platform. Later in 2017 it was believed that this proportion had increased. In December 2017 a random sample of 59 adults over 65 years of age in the UK was collected. It was found that 46 of the 59 adults used this online social media platform.

Use a suitable hypothesis test to determine whether there is evidence at the 1% level to suggest that the proportion of adults over 65 in the UK who used this online social media platform had increased from May 2017 to December 2017. **[7]**

25 Every morning before breakfast Laura and Mike play a game of chess. The probability that Laura wins is 0.7. The outcome of any particular game is independent of the outcome of other games. Calculate the probability that, in the next 20 games,

(a) Laura wins exactly 14 games, **[2]**

(b) Laura wins at least 14 games. **[2]**

26 The screenshot in Fig. 10 shows the probability distribution for the continuous random variable X, where $X \sim N(\mu, \sigma^2)$.

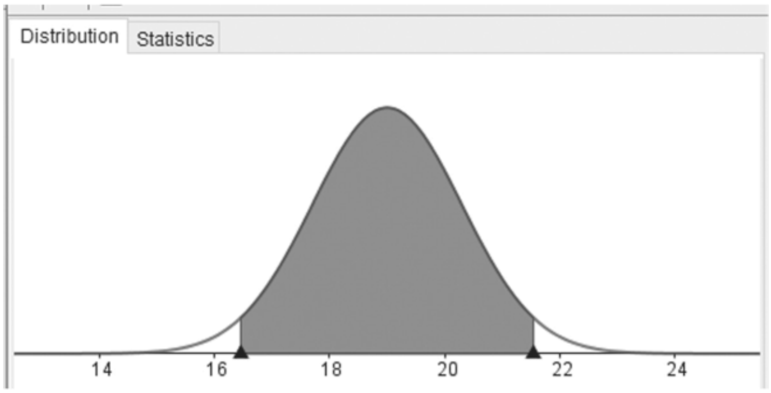


Fig. 10

The area of each of the unshaded regions under the curve is 0.025. The lower boundary of the shaded region is at 16.452 and the upper boundary of the shaded region is at 21.548.

(a) Calculate the value of μ. **[1]**

(b) Calculate the value of σ^2. **[3]**

(c) Y is the random variable given by $Y = 4X + 5$.

(i) Write down the distribution of Y. **[3]**

(ii) Find P(Y > 90). **[1]**

27 The discrete random variable X takes the values 0, 1, 2, 3, 4 and 5 with probabilities given by the formula

$$P(X = x) = k(x + 1)(6 - x) .$$

(a) Find the value of k. **[2]**

In one half-term Ben attends school on 40 days. The probability distribution above is used to model X, the number of lessons per day in which Ben receives a gold star for excellent work.

(b) Find the probability that Ben receives no gold stars on each of the first 3 days of the half-term and two gold stars on each of the next 2 days. **[2]**

(c) Find the expected number of days in the half-term on which Ben receives no gold stars. **[2]**

28 Each weekday Keira drives to work with her son Kaito. She always sets off at 8.00 a.m. She models her journey time, x minutes, by the distribution $X \sim N(15,4)$.

Over a long period of time she notes that her journey takes less than 14 minutes on 7% of the journeys, and takes more than 18 minutes on 31% of the journeys.

(a) Investigate whether Keira's model is a good fit for the data. **[3]**

Kaito believes that Keira's value for the variance is correct, but realises that the mean is not correct.

(b) Find, correct to two significant figures, the value of the mean that Keira should use in a refined model which does fit the data. **[2]**

Keira buys a new car. After driving to work in it each day for several weeks, she randomly selects the journey times for n of these days. Her mean journey time for these n days is 16 minutes. Using the refined model she conducts a hypothesis test to see if her mean journey time has changed, and finds that the result is significant at the 5% level.

(c) Determine the smallest possible value of n. **[5]**

29

The function $f(x) = \dfrac{e^x}{1 - e^x}$ is defined on the domain $x \in \mathbb{R}$, $x \neq 0$.

(i) Find $f^{-1}(x)$. **[3]**

(ii) Write down the range of $f^{-1}(x)$. **[1]**

30 The discrete random variable X takes the values $r = 0,1,2,3,4$ with probabilities

$$P(X = r) = k(r + 1)(r + 2).$$

(a) Calculate the value of the constant k. **[2]**

(b) Calculate $P(X < 3)$, giving your answer as a fraction in its lowest terms. **[2]**

31 **In this question you must show detailed reasoning.**

Each evening Statto goes for a walk on the same circular route. Over a long period of time Statto noticed that the mean time taken to complete the walk was 56 minutes and the standard deviation was 4 minutes.

A few months ago Statto was ill and was unable to complete the walk for a month. Since then Statto's partner thinks that the mean time Statto takes to complete the evening walk has increased. Over a period of weeks Statto's partner collects a random sample of the times taken by Statto to complete the evening walk.

Statto's partner uses software to generate the summary statistics in Fig. 14.

Statistics	
n	19
Mean	57.4737
σ	3.8848
s	3.9912
Σx	1092
Σx^2	63048
Min	52
Q1	54
Median	57
Q3	60
Max	65

Fig. 14

Use information from Fig. 14 to conduct a hypothesis test to determine whether there is any evidence at the 5% level to suggest that the mean time Statto takes to complete the evening walk has increased. **[7]**

32 Rose packs eggs in boxes of 6, which she then sells at her farm. During the process some eggs are cracked, and Rose randomly selects a sample of boxes and records the number of cracked eggs in each box. The results are summarised in Fig. 16.

Number of cracked eggs	0	1	2	3	4	5	6
Number of boxes	163	103	28	5	0	0	1

Fig.16

(a) Calculate the mean number of cracked eggs per box. **[1]**

Rose believes that the number of cracked eggs per box may be modelled by a binomial distribution.

(b) State a modelling assumption that is necessary for a binomial distribution to be used to model the number of cracked eggs per box. **[1]**

Rose defines p as the probability that a particular egg is cracked.

(c) Use your answer to part (a) to find the value of *p*. **[2]**

(d) Calculate the expected frequencies of cracked eggs per box according to Rose's model, giving your answers correct to 1 decimal place. **[3]**

(e) Comment on whether Rose's model is a good fit for the data. **[1]**

Instead of selling eggs at the farm, Rose decides to sell them to a wholesaler. The eggs are now selected randomly and packed in open trays of 24. Rose believes that this will result in a change in the probability of an egg being cracked. She selects a tray at random and finds that 5 eggs are cracked.

(f) In this question you must show detailed reasoning.

Conduct a hypothesis test to determine whether there is any evidence at the 5% level to suggest that the proportion of cracked eggs has changed. **[7]**

33 Two siblings, Layla and Wilson, have identical smart phones with an app called “Screen Time” which provides the user with statistics concerning the amount of time spent using the screen per day. The statistics for several months of use for Layla’s phone are shown in Fig. 8. Time is measured in minutes.

Number of days	120
Shortest time	12.2
Lower Quartile	46.4
Median	56.1
Upper Quartile	66.1
Mean	56.2
Standard deviation	14.5
Longest time	99.8

Fig. 8

Layla believes that X, the amount of screen time she uses per day on her smart phone, may be modelled by the Normal distribution.

(a) **Without** doing any calculation, state one feature of the data in Fig. 8 which suggests that Layla may be correct. **[1]**

(b) Verify that the values for the quartiles are consistent with the model **[2]**

$X \sim N(56.2, 14.5^2)$.

(c) Calculate $P(X > 90)$. **[1]**

Layla believes that Y, the amount of screen time used per day by Wilson on his smart phone, may be modelled by $Y = 2X$.

(d) Use this model to calculate $P(Y > 90)$. **[2]**

Each night Layla and Wilson leave their phones on the kitchen table before going to bed. On one occasion their father picked up one of the phones and said:

"This phone has been used for more than an hour and a half today. There is a 99% chance that this is Wilson's phone."

(e) Use Layla's models for the distributions of X and Y to determine whether Layla and Wilson's father's statement is correct. **[2]**

34 Roxanne is employed by an employment agency called Supertemps. In any given week she may be required to work up to a maximum of 5 days.

Both Roxanne and her partner, Alex, suggest ways in which the number of days, X, Roxanne is required to work in a week may be modelled.

- Roxanne thinks that X may be modelled using the discrete uniform distribution.
- Alex suggests the following formula to model the probability distribution of X.

$$P(X = x) = k(2x - 5)^2 \text{ for } x = 0, 1, 2, 3, 4, 5.$$

(a) Calculate the value of k for Alex's model. **[2]**

Alex and Roxanne record the number of days Roxanne is required to work in each of the first 42 weeks of her employment. The results are summarised in Fig. 9.

Number of days	0	1	2	3	4	5
Frequency	13	7	1	0	7	14

Fig. 9

(b) Investigate how well each model fits the data. [6]

35 For a period of 20 years until the end of 2017 the parish council of the village of Springmoor have met every fortnight to discuss local affairs. Throughout this period Mr. Loparts acted as chairperson.

The parish council secretary has noted that the mean length of time of meetings throughout this period was 142 minutes, and the standard deviation of the length of time of meetings was 17.4 minutes.

At the beginning of 2018 Miss Patel took over as chairperson. At the end of 2018 the secretary stated that he believed that the mean length of time of meetings is lower since Miss Patel became chairperson.

The secretary generated a random sample of lengths of times of meetings in 2018. The times, in minutes, are as follows.

123, 137, 119, 135, 136, 143, 126, 126, 125.

Assume that the length of time of meetings may be modelled by the Normal distribution and that the variance of the length of time of meetings is unaltered.

Conduct a hypothesis test at the 5% level to determine whether there is any evidence to support the secretary's belief that the mean length of meeting has decreased.

[7]

36 The 24 members of staff at the Blackley branch of the Midshires Bank decide to start a lottery. They will each pay £5 per week to enter the lottery.

Each week, three of the numbers 1, 2, 3 and 4 will be placed in a random order by a computer. Each member of staff will make 3 different attempts, entered into their computer workstation, to guess what the computer has chosen.

(a) Show that the probability that a particular member of staff enters the correct numbers in the correct order in any week is $\frac{1}{8}$. **[3]**

If exactly one person chooses the right numbers in the right order, that person will win a prize of all £120 paid by staff to enter the lottery. If **no-one** chooses the right numbers, or **if more than one person** chooses the right numbers, the prize will be 'rolled over' to the next week.

(b) Calculate the probability that someone will win the prize in the first week. **[2]**

(c) Calculate the probability that the customer services manager, Marak, will win the prize in each of the first two weeks. **[2]**

In fact, when the lottery started, nobody won until the prize reached £6000. At that point Milena, the chief cashier, won. Marak commented that there was a one in a million chance of this happening to Milena.

(d) Determine whether Marak's comment is correct. **[4]**

37(a) Fig. 1 shows the probability distribution of the discrete random variable X.

x	1	2	3	4	5
$P(X = x)$	0.2	0.1	k	$2k$	$4k$

Fig. 1

Find the value of k. **[2]**

(b) Find $P(X \neq 4)$. **[2]**

38(a) A team called “The Educated Guess” enter a weekly quiz. If they win the quiz in a particular week, the probability that they will win the following week is 0.4, but if they do not win, the probability that they will win the following week is 0.2.

In week 4 The Educated Guess won the quiz.

Calculate the probability that The Educated Guess will win the quiz in week 6. **[3]**

(b) Every week the same 20 quiz teams, each with 6 members, take part in a quiz. Every member of every team buys a raffle ticket. Five winning tickets are drawn randomly, without replacement. Alf, who is a member of one of the teams, takes part every week.

Calculate the probability that, in a randomly chosen week, Alf wins a raffle prize. **[2]**

(c) Find the smallest number of weeks after which it will be 95% certain that Alf has won at least one raffle prize. **[4]**

39(a) Club 65–80 Holidays fly jets between Liverpool and Magaluf. Over a long period of time records show that half of the flights from Liverpool to Magaluf take less than 153 minutes and 5% of the flights take more than 183 minutes.

An operations manager believes that flight times from Liverpool to Magaluf may be modelled by the Normal distribution.

Use the information above to write down the mean time the operations manager will use in his Normal model for flight times from Liverpool to Magaluf. **[1]**

(b) Use the information above to find the standard deviation the operations manager will use in his Normal model for flight times from Liverpool to Magaluf, giving your answer correct to 1 decimal place. **[3]**

(c) Data is available for 452 flights. A flight time of under 2 hours was recorded in 16 of these flights. Use your answers to parts **(a)** and **(b)** to determine whether the model is consistent with this data. **[3]**

(d) The operations manager suspects that the mean time for the journey from Magaluf to Liverpool is less than from Liverpool to Magaluf. He collects a random sample of 24 flight times from Magaluf to Liverpool. He finds that the mean flight time is 143.6 minutes.

Use the Normal model used in part **(c)** to conduct a hypothesis test to determine whether there is evidence at the 1% level to suggest that the mean flight time from Magaluf to Liverpool is less than the mean flight time from Liverpool to Magaluf. **[7]**

(e) Identify two ways in which the Normal model for flight times from Liverpool to Magaluf might be adapted to provide a better model for the flight times from Magaluf to Liverpool. **[2]**

40 **You must show detailed reasoning in this question.**

The screenshot in Fig. 15 shows the probability distribution for the continuous random variable X, where $X \sim \mathrm{N}(\mu, \sigma^2)$.

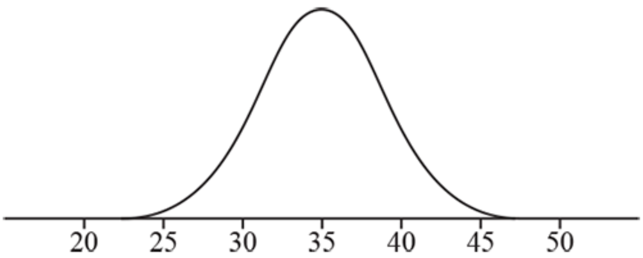


Fig. 15

The distribution is symmetrical about the line $x = 35$ and there is a point of inflection at $x = 31$.

Fifty independent readings of X are made. Show that the probability that at least 45 of these readings are between 30 and 40 is less than 0.05. **[6]**

41(a) According to the latest research there are 19.8 million male drivers and 16.2 million female drivers on the roads in the UK.

A driver in the UK is selected at random. Find the probability that the driver is male. **[1]**

(b) Calculate the probability that there are 7 female drivers in a random sample of 25 UK drivers. **[1]**

(c) When driving in a built-up area, Rebecca exceeded the speed limit and was obliged to attend a speed awareness course. Her husband said “It’s nearly always male drivers who are speeding.” When Rebecca attends the course, she finds that there are 25 drivers, 7 of whom are female. You should assume that the drivers on the speed awareness course constitute a random sample of drivers caught speeding.

In this question you must show detailed reasoning.

Conduct a hypothesis test to determine whether there is any evidence at the 5% level to suggest that male drivers are more likely to exceed the speed limit than female drivers. **[7]**

(d) State a modelling assumption that is necessary in order to conduct the hypothesis test in part **(c)**. **[1]**

42(a) In a certain country it is known that 11% of people are left-handed.

Calculate the probability that, in a random sample of 98 people from this country, 5 or fewer are found to be left-handed, giving your answer correct to 3 significant figures. **[1]**

(b) An anthropologist believes that the proportion of left-handed people is lower in a particular ethnic group.

The anthropologist collects a random sample of 98 people from this particular ethnic group in order to test the hypothesis that the proportion of left-handed people is less than 11%.

The anthropologist carries out the test at the 1% level.

Determine the critical region for this test. **[3]**

43(a) Rosella is carrying out an investigation into the age at which adults retire from work in the city where she lives. She collects a sample of size 50, ensuring this comprises of 25 randomly selected retired men and 25 randomly selected retired women.

State the name of the sampling method she uses. **[1]**

(b) Fig. 8.1 shows the data she obtains in a frequency table and Fig. 8.2 shows these data displayed in a histogram.

Age in years at retirement	45 –	50 –	55 –	60 –	65 –	70 –	75 – 80
Frequency Density	0.4	1.8	2.4	2.2	1.8	1.2	0.2

Fig. 8.1

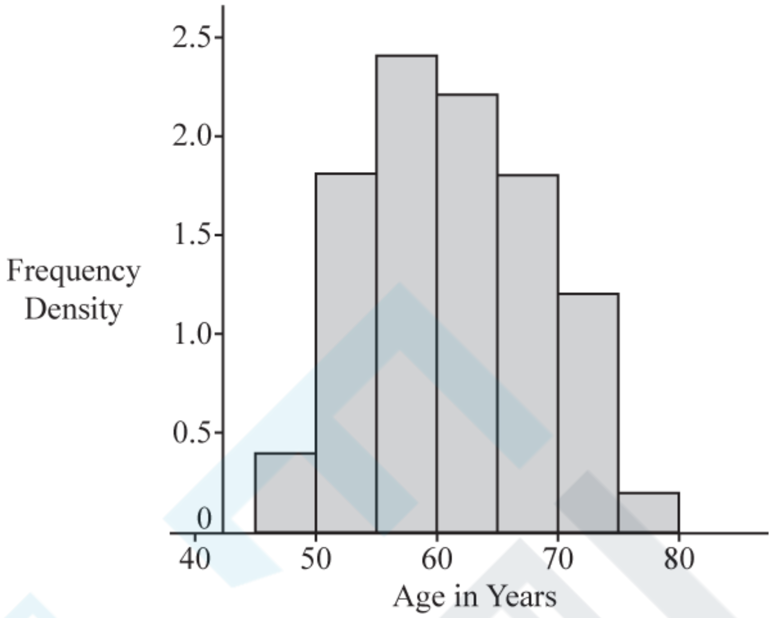


Fig. 8.2

How many people in the sample are aged between 50 and 55? **[1]**

(c) Rosella obtains a list of the names of all 4960 people who have retired in the city during the previous month.

Describe how Rosella could collect a sample of size 200 from her list using

- systematic sampling such that every item on the list could be selected,
- simple random sampling. **[4]**

(d) Rosella collects two simple random samples, one of size 200 and one of size 500, from her list. The histograms in Fig. 8.3 show the data from the sample of size 200 on the left and the data from the sample of 500 on the right.

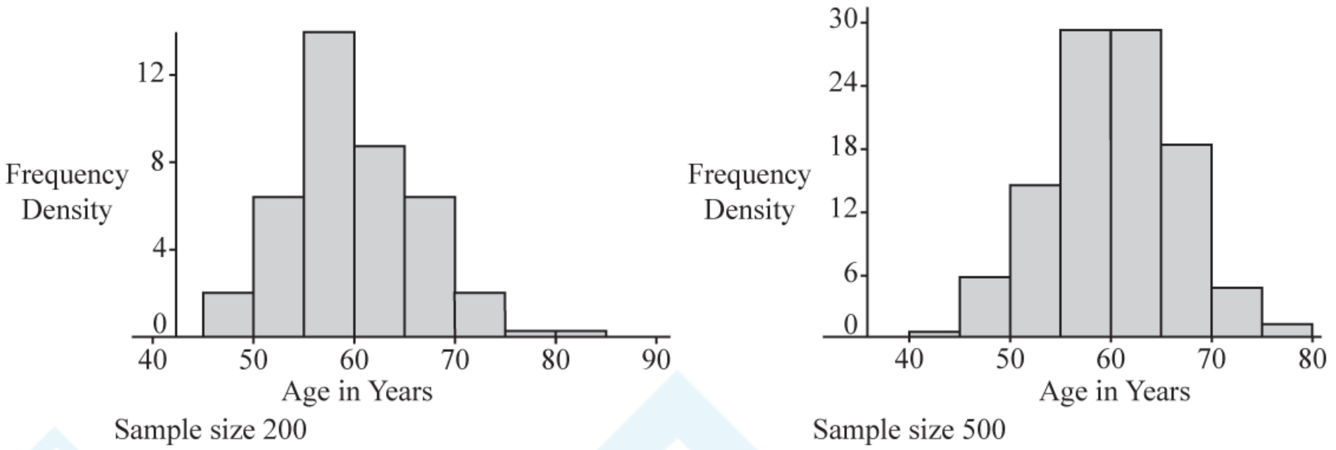


Fig. 8.3

With reference to the histograms shown in Fig. 8.2 and Fig. 8.3, explain why it appears reasonable to model the age of retirement in this city using the Normal distribution. **[1]**

(e) Summary statistics for the sample of 500 are shown in Fig. 8.4.

Statistics	
n	500
Mean	60.0515
σ	6.5717
s	6.5783
Σx	30025.7601
Σx^2	1824686.322
Min	36.0793
Q1	55.2573
Median	59.9202
Q3	64.4239
Max	81.742

Fig. 8.4

Use an appropriate Normal model based on the information in Fig. 8.4 to estimate the number of people aged over 65 who retired in the city in the previous month. **[4]**

(f) Identify a limitation in using this model to predict the number of people aged over 65 retiring in the **following** month. **[1]**

44 A company supplies computers to businesses. In the past the company has found that computers are kept by businesses for a mean time of 5 years before being replaced. Claud, the manager of the company, thinks that the mean time before replacing computers is now different.

Claud decides to conduct a hypothesis test at the 5% level to test whether there is evidence to suggest that the mean time that businesses keep computers is not 5 years. He takes a random sample of 120 computers. Summary statistics for the length of time computers in this sample are kept are shown in Fig. 9.

Statistics	
n	120
Mean	4.8855
σ	2.6941
s	2.7054
Σx	586.2566
Σx^2	3735.1475
Min	0.1213
Q1	2.5472
Median	4.8692
Q3	7.0349
Max	9.9856

Fig. 9

In this question you must show detailed reasoning.

• State the hypotheses for this test, explaining why the alternative hypothesis takes the form it does.
• Use a suitable distribution to carry out the test. **[8]**

45(a) **In this question you must show detailed reasoning.**

A 5-sided spinner can give scores of 1, 2, 3, 4 or 5. After observing a large number of spins, Elaine models the probability distribution of X, the score on the spinner, as shown in Fig. 12.

x	1	2	3	4	5
$P(X = x)$	0.2	0.3	p	p	q

Fig. 12

When the spinner is spun twice, the probability of obtaining a total score of 9 is 0.06.

Given that $q < 2p$, determine the values of p and q. **[6]**

(b) The spinner is spun 10 times. Calculate the probability that exactly one 5 is obtained. **[2]**

(c) Elaine's teacher believes that the probability that the spinner shows a 1 is greater than 0.2. The spinner is spun 100 times and gives a score of 1 on 28 occasions.

Conduct a hypothesis test at the 5% level to determine whether there is any evidence to suggest that the probability of obtaining a score of 1 is greater than 0.2. **[7]**

END OF QUESTION PAPER